ONLINE STORE SALES DATA ANALYTICS PROJECT

Project Report | Prepared by Maheswari S.

1. Introduction

This project analyzes online store sales data using Python. The aim is to understand revenue, product demand, category performance, city-wise sales, and monthly sales trends. The analysis was completed in a Jupyter Notebook using Pandas and Matplotlib.

2. Objectives

- Calculate total revenue from the sales records.
- Identify the product with the highest quantity sold.
- Find the city generating the highest revenue.
- Compare revenue across product categories.
- Review monthly revenue trends and identify stronger and weaker periods.
- Prepare charts that make sales patterns easier to compare.

3. Tools and Technologies

- Python — data analysis and processing
- Pandas — loading, cleaning, transforming, and summarizing data
- Matplotlib — creating sales charts
- Microsoft Excel — source dataset and cleaned output
- Jupyter Notebook — writing and running the analysis

4. Dataset Description

The project uses the Excel dataset sales_data_2000_rows (2) (2).xlsx. The notebook shows 2,000 sales records and the original fields Order_ID, Date, Product, Category, City, Quantity, and Price. Two derived fields were added during analysis: Revenue and Month.

Dataset detail	Value
Records	2,000
Original fields	7
Derived fields	Revenue, Month
Date coverage	January 2025 to January 2026 (January 2026 is partial)

5. Methodology and Data Preparation

- Loaded the Excel file into a Pandas DataFrame and inspected the first rows and dataset information.
- Checked missing values and duplicate rows. The notebook found no duplicate rows.
- The initial missing-value check showed one missing Quantity value. In the final notebook output, missing values were zero after the quantity was handled.
- Converted Date to a datetime data type.
- Calculated Revenue using Quantity × Price.
- Created a Month field from Date to support monthly aggregation.
- Grouped data by product, city, category, and month to compare performance.

6. Analysis and Actual Results

Metric	Notebook result
Total revenue	51,501,934
Highest quantity product	Chair — 568 units
Highest revenue city	Ahmedabad — 7,449,911
Highest revenue category	Electronics — 37,095,222
Lowest revenue city	Bangalore — 5,161,391
Lowest revenue category	Stationery — 157,159
Lowest quantity product	Table — 449 units
Highest monthly revenue	December 2025 — 4,891,185
Next highest monthly revenue	May 2025 — 4,884,091

7. Visualizations Created

- Monthly Sales Trend — line chart showing revenue by month.
- Product-wise Quantity Sold — bar chart comparing units sold by product.
- Revenue by Category — bar chart comparing revenue across categories.

8. Key Findings

- Total revenue across the 2,000 records was 51,501,934.
- Chair had the highest total quantity sold, at 568 units. Table had the lowest, at 449 units.
- Ahmedabad generated the highest city-wise revenue, while Bangalore generated the lowest.
- Electronics was the highest-revenue category by a wide margin; Stationery was the lowest.
- December 2025 had the highest monthly revenue in the notebook output, closely followed by May 2025.
- January 2026 has a much smaller revenue value than the 2025 months and should be treated as a partial month rather than a full-month comparison.

9. Recommendations

- Monitor stock levels for high-demand products such as Chair.
- Review city-wise sales strategies and investigate why Bangalore revenue is lower than Ahmedabad.
- Evaluate product range, pricing, and promotion for lower-revenue categories such as Stationery.
- Plan campaigns using monthly trends, while comparing full months with full months.
- Continue checking missing values and data quality whenever new sales data is added.

10. Conclusion

This project demonstrates a basic data analytics workflow using Python, Pandas, and Matplotlib. The dataset was inspected and prepared, revenue and month fields were created, and sales were compared by product, city, category, and month. The results provide a clear starting point for understanding sales performance and planning business improvements.

11. Project Files

- Source dataset: sales_data_2000_rows (2) (2).xlsx
- Notebook: sales_analysis(1).ipynb
- Cleaned output created by the notebook: cleaned_sales_data.xlsx
- Report: Online_Store_Sales_Analysis_Report_Black_White.pdf